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Chapter 10

Tips And Techniques



Saving And Installing Application Data

The purpose of the Gold Copy process is to enable application developers to create and maintain standard applications after the development process is completed which can be shipped to other processors or be used on the same processor.

An application is defined by its identity code for example, IN - Inventory, GL - General Ledger and its release level. The environment code allows the user to create a series of separate development areas or environments for the same application. It is recommended that each of these is intrinsically linked to the Standard Application which has a blank environment code. So for example, the Inventory module may be set up as follows:

Application Description: Standard Inventory

Application ID: IN

Environment: blank

Release: 02

Application Description: Inventory Test Version

Application ID: IN

Environment: TST

Release: 02

For application development purposes it is essential that different environments can be created and that any new Application Manager data can be merged with the existing standard application information. For example, developing a new function for an existing application may involve the creation of additional tasks and/or menus. These would normally be tested in an alternative environment before being transferred and subsequently released into the live environment.

Gold Copy



Note: Gold Copy can be initiated either by selecting option 6 from the Application Manager Utilities sub menu or by entering the command AMGOLD.

The Gold Copy function is provided to perform two key activities:

Activity 1

To merge Application Manager data that has been set up for either development or test purposes with the standard application data.

The Gold Copy routine will search through Application Manager's files looking for the selected application. The following information needs to be entered in order to select an application; Application ID, Environment and Release. The following data is extracted:

- Exclusivity groups
- Application profile
- Library lists
- Application tasks
- Menus - the Gold Copy function selects the menus to be extracted by first identifying all of the menus that belong to the specified Menu environment and then for each of the menus selected identifying which of those menus belong to the Application specified
- Task search words
- Report manager information.

Each of the extracted elements is then matched with the standard application data. If there is a match then the existing standard information is replaced. If there is no match then the information is added to the standard application. In essence the environment code is changed to blanks.



Note: Menus will probably contain application tasks. If those tasks reference the test or development application which is being Gold Copied, those tasks will be changed to reference standard application tasks. Tasks contained within the menu that reference other applications will not be changed.



Note: The Gold Copy process does not delete data that is no longer required from standard application. You should remove this data manually after the Gold Copy process has completed. For example, the library lists that have merged from your development environment may contain development or test libraries, which should be removed.

When the Gold Copy task has completed you should verify the new standard application using Application Manager's report, enquiry and maintenance functions to ensure all of library lists, application tasks, menus and so on are correct. If you make any amendments during this verification phase then the application must be Gold Copied again.



Note: This phase of the Gold Copy process will only be performed if the environment code is not blank.

Activity 2

The second activity in the Gold Copy process is to extract the standard application data that has been generated and to create Installation Files. The Installation Files contain only the data required to run the standard application. The Installation Files are used by Application Manager's installation routines when an application is to be loaded on another processor. The Installation Files will be created in the library specified when the Gold Copy process was initiated. It is recommended that the Installation Files are stored in the Applications Program Library.



Note: The installation files are given names with the following format: APGnnI where nn represents the corresponding Application Manager file named APGnnPHY.



Note: If you wish to create the Installation Files and not merge Application Manager data then use the command AMENVGOLD. This command is particularly useful if you wish to install a bespoke or custom application on another Server.

Application Installation

The command AMINSTALL in library IPGAMP4 is provided to perform the installation of Application Manager data on another system. You must have already restored the applications libraries and the Installation Files which will normally be held in the applications program library.

The Application Installation process performs the same application merging as the Gold Copy routine as described in the Using Menus for Command Entry section. In addition, when the first application is installed on a processor it will also perform the following:

Prompts for and then sets the Users Default Environment code, which is held in Data Area AMCUSCODE in library IPGAMP4. Once you have set your default environment code it may not be changed.



Note: The only exception to this rule is if you have performed an AMINSTALL using environment code OSL, you may change your code at the next install. This will have the same effect as a first installation and will clear out library lists, tasks, menu's etc.

Clears all Application Manager files in library IPGAMF3. This is because Application Manager is normally shipped with Demonstration applications.

Creates the following menus:

- Initial Menu-AM0
- Company Profiles Menu-AM1
- Fast Path Menu Creation Standard Header- @STDMNU

For each application installed the user profiles QPGMR and QUSER have User Profiles created or updated to enable them to have authorisation to all of the tasks and menus that the application contains. In addition they are also authorised to company ZM for each of the applications installed. See the Administration Functions Product Guide for more information about user profiles.



Note: If a library is detected in a library list that does not exist on the processor on which the application is being installed, then that library will be automatically removed from the application's library list. Therefore, it is important that you have restored all of the libraries required by interfacing applications before you run the Application Manager installation routine.



Note: If you are installing applications over a period of time or have a phased approach to your implementation, it is important that for applications already installed, you either change their library lists manually to reflect the installation of new interfacing applications, or you

perform the Installation process again for those applications.

Using Menus For Command Entry

Application Manager menus allow commands to be entered as well as allowing the user to select menu options.



Note: Commands can only be entered if the User has sufficient authority. Authority to enter commands is controlled by the Administration Functions User Profile: see the Maintain User Profiles section in the Administration Functions Product Guide.

This means that users are allowed to enter IBM OS/400, JBA or your own bespoke commands. The Application Manager module main task is to be the driver of your applications to do this it makes extensive use of library lists and the LDA (local data area). If the user accesses function by entering commands then Application Manager will not protect (or save) the library list or the LDA. Thus the library list and LDA will not be saved before execution of a command, and restored back, when the user completes the execution of the command and returns back to the control of Application Manager.



Note: Application Manager does not save the library list and LDA before the execution of commands, as this is a time-consuming activity, and would impact the run time performance of Application Manager.

Therefore if you are going to execute commands from Application Manager menus that affect the library list and/or the LDA, you should save them before executing the command and restore them back afterwards. There are a number of commands supplied which enable you to do this they are:

- PUTLDA
- GETLDA
- PUTLIBL
- GETLIBL.



Note: The majority of IBM OS/400 commands have no effect on either the library list or the LDA and thus can be executed without causing any problems.

The Local Data Area LDA

Application Manager makes extensive use of the Local Data Area LDA in order to pass information between programs. The module has a fixed and pre-defined view of the first 256 characters of the 1024 character LDA. The remainder of the Local Data Area is specifically for application usage. You must ensure that your applications never change the first 256 characters of the Local Data Area.



Note: The exception to this rule is the return code L#RTCD positions 118 to 119 of the Local Data Area. If you are changing the return code then you save the LDA before you execute your routine and restore it back when your routine has completed.

Command List

This section lists the commands that are available to you from the Administration Function product.

CLNUSRALC : Cleanup user allocations

The command CLNUSRALC will clear out any user allocations that are no longer valid. User allocations are used to determine the number of users of an application. It maybe the case that a job was terminated abnormally and the user is still allocated to an application.

This command may be run at anytime.

The command has no parameters.

See the section User Code Monitoring for further details.

DSPAUTERR : Display authorisation violations

The command DSPAUTERR is used to display the Authorisation Violation Log. See the section Display Violation Log for further details.

DSPCTLDTA : Display application control data

The command DSPCTLDTA is to display Gold Copy details about the products. To display details about a particular product enter DSPCTLDTA aaaaaaaaaa.



Note: Where aaaaaaaaaa is a library name.

FINISH : ENDGRPJOB, ENDPASTHR or SIGNOFF

The command FINISH has the effect of closing down any group jobs that are active for the workstation session. In addition if display station pass through is in operation the session will be cleared down and your interactive job on the target Server will be signed off. If the session is not using group jobs or pass through then the session will simply be signed off.

MNTIPGLIBL : Maintain Internal Library List

The command MNTIPGLIBL allows you to modify the library lists that will be used by the System Manager modules and the JBA development products.

The modules are identified by the following two character codes:

- AM Application Manager.
- CF Manager 400 Sign on screen.
- FM File Migration Utility.
- HK Housekeeping.
- MM Machine Manager.
- NW Network Manager.
- R# IRG/400 (Report Generator).
- AF Administration Functions.
- V# VIEWfinder.

Used in conjunction with Language Code, it provides the mechanism of having many language versions of the same module on a single processor. For further information see sections entitled Maintain User Profiles and Maintain Language Parameters.

RUNDWL : Run VIEWfinder extract

The command RUNDWL is used to run data extracts created by VIEWfinder. For further details about this command please refer to the VIEWfinder Data Warehouse Loader documentation.

RUNRGREP : Run IRG/400 Report

The command RUNRGREP is used to run report programs created by IRG/400 (Report Generator). For further details about this command please refer to the IRG/400 documentation.

SETIPGLIBL : Set Internal Library List

The command SETIPGLIBL allows you to change your library list to contain the libraries from one of the following sources:

- System Manager modules.
- Development products.
- Applications controlled by Application Manager.

For details about how the contents of library lists are maintained see the command MNTIPGLIBL and refer to the Application Manager Product Guide.

The command SETIPGLIBL has the following parameters:

- Application ID (APPL).
- Version (VRSN) Not used if the Library list type is *IPG.
- Release (RLSL) Not used if Library list type is *IPG.
- Library List Type (TYPE) Defines how library list details are obtained. The allowed values are:
 - IPG indicates use the library list defined using the command MNTIPGLIBL
 - *USER indicates use the application's library list maintained via IPG/400, IRG/400, VIEWfinder and Application Manager using the library list code *DFT
 - *MGR indicates use the application's library list maintained via Application Manager, using the application's default library list code.
 - Replace Existing LIBL (REPLACE) Valid values are *YES and *NO
 - Position in LIBL (POSITION) Only used if you specify Replace Existing LIBL *NO. Valid values are *FIRST and *LAST.



Note: The user's language code will also be used to determine which library list to use.

SETV2 : Set version two library list

The command SETV2 may be used to change your system library list.

It changes from a Version 3 to a Version 2 environment.

You may prompt the command to change the From and To libraries.



Note: This command is only relevant to users who have Versions 2 and 3 of System Manager.

SETV3 : Set version three library list

The command SETV3 may be used to change your system library list.

It changes from a Version 2 to a Version 3 environment.

You may prompt the command to change the From and To libraries



Note: This command is only relevant to users who have Versions 2 and 3 of System Manager.

SMISYSCMDS : Install System Library Commands

The command SMISYSCMDS allows you to populate the library to be your System Library with objects from one or all of the System Manager and Developer products.

The command has two parameters:

- System Library Name (LIB) The name you have selected for the System Library. The default is OSLSYS.
- Install commands from module (MODULE) A two character module ID may be entered. Leave this parameter blank to indicate all modules.

STRIPGAG : Start IPG/400

The command STRIPGAG may be used to go straight to IPG/400 without the need to use the command STRM400.

The command has three parameters. Application ID (APPL), Version (VRSN) and Release (RLSL). If you specify these parameters you will go directly to that application bypassing the select application screen.

STRIPGAM : Start Application Manager

The command STRIPGAM may be used to go straight to Application Manager without the need to use the command STRM400.

The command has three parameters. Application ID (APPL), Version (VRSN) and Release (RLSL). You may specify to set the default application to maintain.

STRIPGCF : Start Administration Functions

The command STRIPGCF may be used to go straight to the Common Functions module without the need to use the command STRM400.

STRIPGHK : Start Housekeeping

The command STRIPGHK may be used to go straight to the Housekeeping module without the need to use the command STRM400.

STRIPGMM : Start Machine Manager

The command STRIPGMM may be used to go straight to the Machine Manager module without the need to use the command STRM400.

STRIPGNW : Start Network Manager

The command STRIPGNW may be used to go straight to the Network Manager module without the need to use the command STRM400.

STRIPGRW : Start IRG/400 – Report Generator

The command STRIPGRW may be used to go straight to IRG/400 (Report Generator) without the need to use the command STRM400.

The command has three parameters. Application ID (APPL), Version (VRSN) and Release (RLSL). If you specify these parameters you will go directly to that application bypassing the select application screen.

STRIPGVF : Start VIEWfinder

The command STRIPGVF may be used to go straight to VIEWfinder without the need to use the command STRM400.

The command has three parameters. Application ID (APPL), Version (VRSN) and Release (RLSL). If you specify these parameters you will go directly to that application bypassing the select application screen.

STRM400 : Start System Manager

The command STRM400 is used to display the System Manager main menu. This menu allows access to all of the System Manager and Developer modules.



Note: See the section entitled System Manager for further details.

Company Validation Program

The majority of multi-company applications will include a company profile file. This file will hold company level information. When processing a multi-company application it will normally be necessary to check for the existence of the company being processed on the company profile file.

When a task is executed by Application Manager, the menu driver does not know the name or format of an applications company profile. Therefore it is recommended that a program is written that will check for the existence of the company selected to process. This program will be automatically called when you access a task code which requires company validation. The program to call is described in the Application Profile, see the Maintain Applications topic.



Note: Only task codes between 0100 and 8999 will perform company validation checking, see the Maintain Tasks topic.

Your validation program will need to perform the following function:

If the company selected exists set the LDA (Local Data Area) field L#RTCD (Return Code) to blanks. Application Manager will then allow the task the user has selected to be executed.

If the company selected does not exist set the LDA (Local Data Area) field L#RTCD (Return Code) to NC. Application Manager will not initiate the task the user has selected but will instead return them to the menu and display the message 'This company does not exist in the application requested.'



Note: The return code L#RTCD is contained in positions 118 to 119 of the Local Data Area.



Note: Your company validation program will need to determine which company the user has selected the LDA field L#CONO - company number contains the company the user has selected. The field L#CONO is contained within positions 1 and 2 of the Local Data Area.

Change Company

When a user enters the Application Manager menu environment, the default company to process is taken from the Administration Functions User Profile. If the user has a default company, then this will be made the default for all applications.

The JBA change company command CHGCO in its simplest form is used to change the default company across all applications.



Note: Standard menus are provided with function key 16 set up to execute the CHGCO command.

In situations where the user has no default company assigned on their Administration Functions User Profile he will be automatically prompted to select the company required the first time he enters each application. Thus a different company code may be selected for each application the user has entered. For example:

CHGCO nn

Where nn is the company code you wish to use.

Tasks With Interactive And Batch Programs

An application task may have to execute both an interactive and a batch program. For example, a report may require selection parameters to be completed before the report program is submitted for processing.

The interactive program processed before the report program is submitted may allow the user to decide whether the report should be run. If the user decides not to run the report, then this information should be passed to Application Manager, to prevent the report program being submitted. To do this the interactive program should do the following:

If the batch program is to be submitted then set the LDA (Local Data Area) field L#RTCD (Return Code) to Y. If the return code is not set to Y the batch job will not be submitted.



Note: The return code L#RTCD is contained in positions 118 to 119 of the Local Data Area.

Language Codes

The various languages you may wish to use are set up within Administration Functions (see Maintain Language Parameters). The Administration Functions User Profile allows language codes to be set at user level.

The Application Profile (see the Maintain Applications section) holds a four-character default library list code; this is used if the user's language code on the Administration Functions User Profile is blank.

If the user's language code is not blank, then the default library list code is constructed in the following way:

The first two characters are made up from the user's language code.

The last two characters are taken from the last two characters of the default library list code from the Applications Profile.

For example, a user may have a language code of FR-France and is using an application with a default library list code set up as 0001. The library list to be used will be FR01.



Note: If the library list is specified at task level, thus overriding the default application library list, then the library list code is constructed in the same manner as described above, except that the last two characters are taken from the library list code specified at task level.

Environment Usage

When users enter Application Manager's menu driver they can sign on and use one of two generic environments:

Environment 1

The normal default environment (their personal environment code). This is determined by the Data Area `AMCUSCODE` in library `IPGAMP4`. The user can enter the normal environment in a number of ways.

- By entering the command `IPGAMP4/AM3`
- By selecting either option 1 or 2 from the Application Manager Execute applications sub menu.
- By having the program `IPGAMP4 /AMINITIAL` as the user's Initial Program To Call, so that when they sign on to the Server, they directly enter Application Managers menu driver.

Environment 2

An alternative environment; for example, one used for testing or training purposes. The user can enter the alternative environment in a number of ways:

- By entering the command `IPGAMP4/AMT`
- By selecting either option 3 or 4 from the Application Manager Execute applications sub menu.
- If you wish users to sign on directly to an alternative environment, you must write a CL program which overrides your system's default environment code. The program should include the following statements:

PGM

```
CRTDTAARA DTAARA(QTEMP/IPGENV) TYPE(*CHAR) LEN(3) +  
VALUE('XYZ')
```

AM3

ENDPGM

where `XYZ` is the environment code to be used during the session. This program should then be the user's initial program specified on their Server's user profile.

Application Manager will initially access Menus, Tasks and Library Lists using the selected environment code. However, there will be occasions when the required record does not exist in these instances Application Manager may use a drop through technique in accessing the record to use.

For example, a test environment may be the same in all respects as the live environment except that the library list used contains test files libraries instead of the application live files. In this case the only difference between the two applications are the Library Lists and the Application Profile. The Menus and Tasks are exactly the same. Application Manager has been designed to cater with the requirement to have alternative environments.

The following tables detail the way in which Application Manager determines the environment code to use when accessing Menus, Tasks and Library Lists.

Dft

The default environment code. The personal code or customer code you were given when installing.

Tst

The environment code used for testing purposes.

Other

Any environment code which is different from both your DEFAULT (DFT) code and your TEST (TST) code if you are using one.

Blank

Standard application data has a BLANK environment code.

Menus

When a user attempts to access a menu they will either be signed on using their default environment code or a test code. The following table shows how a drop-through technique is used to find the required menu. The following tables show the drop through techniques used for menus, tasks and library lists.

User Env.	
DFT	DFT BLANK
TST	TST BLANK

Tasks

For each menu option which executes an application task the Application I.D./Environment/Release code are specified. The environment may be blank, be the same as the operator is using (default), or be something else (other). The following table shows the way in which Application Manager will try and access the task code using the drop-through technique.

User Env.	Task Environment		
	BLANK	DFT	OTHER
DFT	DFT BLANK	DFT	OTHER
TST	TST BLANK	TST DFT	TST OTHER

Library List

Once a task has been chosen, according to the above rules, then the library list to be used will be determined as described in the following table:

User Env.	Library List Environment			
	BLANK	DFT	OTHER	TST
DFT	DFT BLANK	DFT BLANK	OTHER BLANK	-
TST	TST	TST	TST	TST

This gives the following combinations of task/library list environment.

For A DFT User

Menu Task Environment		Application Task Environment	Application Library List
1.	Menu Task BLANK	DFT exists, use DTF	DTF exists use DTF No DFT, use BLANK
	Menu Task BLANK	No DFT, use BLANK	DTF exists use DTF No DFT, use BLANK
2.	Menu Task DFT	DFT	DTF exists use DTF No DFT, use BLANK
3.	Menu Task OTHER	DFT	OTHER exists use OTHER No OTHER, use BLANK

For A TST User

Menu Task Environment		Application Task Environment	Application Library List
1.	Menu Task BLANK	TST exists, use TST No TST, use BLANK	TST TST
2.	Menu Task DFT	TST exists, use TST No TST, use DFT	TST TST
3.	Menu Task TST	TST exists, use TST No TST, use OTHER	TST TST
4.	Menu Task OTHER	TST exists, use TST No TST, use OTHER	TST TST



Note: The menu enquiry (see the Menus section) provides an excellent method of finding out which tasks and library list code will be used.

Menu Literals

For the purpose of Language Translation or to meet your installation's own standards you may wish to change the standard text or literals which are contained in menus.

The majority of these are held in a message file.



Note: Changing the message file will only affect new menus which you create. It will not affect existing menus.



Note: The message file is called XAMSGF

1 - Function keys 1 to 24 are described by message numbers UX A3051 to UX A3074. The first 18 characters contain the key description, the remaining characters the command to execute.

3 - The literal Option is held in message number UX A3075.

4 - The literal Select is held in message number UX A3076.

Submitting Batch Jobs To The Machine Manager Scheduler

Application task batch jobs can either be submitted to a batch queue immediately or, if the module is installed, queue on the Machine Manager Job Scheduler for running later in the day or overnight.

To do this, the user will need to be authorised to prompt on commands on the user profile. This enables the menu function key with the special word Prompt behind it. If the user enters the option number of the report on the required menu and then presses F4 rather than Enter, then a prompt is displayed to change the job queue and output queue parameters of the job to be submitted. The user can also enter a time to execute from the Machine Manager Scheduler.

Program Dumps

If a program terminates abnormally and produces a dump, either from Application Manager itself or an Application then the listing will be placed on an output queue. The queue to be used is controlled by data area XADMPOUTQ in library IPGAMP4. This is initially set to be OSLDUMPS but may be changed according to your own requirements. You must ensure that the library containing this output queue is in your library list at all times whilst you are using Application Manager.

Executing Tasks Externally

The command EXCAMTASK provides the facility to run Application Manager tasks from outside the Application Manager menu handler. It may also, if required, be used to run such tasks even from within the menu handler. The command has the following parameters:

Application Code (APPL)

Specifies the Application to which the tasks belongs.

Application Code

The two-character identifier for the application.

Enter *SELECT to display a select window from which a valid application may be chosen.



Note: This is invalid when EXCAMTASK runs in a batch environment.

Application Environment

The three-character environment code for the application.

Release Level

The two-character code specifying the Release Level of the application.

Application Task (TASK)

Specifies the 4 character code that identifies the application task.

Enter *SELECT to display a select window from which a valid task code may be chosen.



Note: This is invalid when EXCAMTASK runs in a batch environment.

Task Search Word (SCHWRD)

Specifies a search word to be used in searching for the required task.



Note: If search a word is entered, all entries in the Application Code (APPL) and Application Task (TASK) are ignored.

This parameter may not be used when EXCAMTASK runs in a batch environment.

User Environment (USERENV)

***Dft**

Use the default customer code. This is equivalent to using options 1 or 2 from the Execute Applications sub-menu.

User Environment

Specify another environment code. This is equivalent to using options 3 or 4 from the Execute Applications sub-menu in Application Manager.

Company Code (CONO)

The two-character company code used by the application task to process company-specific data.



Note: Any value for CONO will be ignored if the task is not company-validated (that is, the task code value is less than 0100 or greater than 9899).

Enter *SELECT to display a selection window from which a valid Company Code may be chosen.



Note: This is invalid when EXCAMTASK runs in a batch environment.

User Profile (USER)

***Current**

The current user's user profile will be in force while the command is running. Company and task authorities will be checked against the current user.

Enter *SELECT to display a selection window from which a valid user profile may be chosen.



Note: This is invalid when EXCAMTASK runs in a batch environment.

User Profile Name

Specify a valid user profile to be used while the task is being executed. This allows more than one user to make use of a single user's company and task authorities. Any job queue and output queue overrides defined for the specified user will also be implemented.



Note: This does not affect system object authorities, only Application Manager task and company authorities.

Run Recovery Program If Needed (RUNRECOVER)

***NO**

Do not run the task's recovery program if the task terminated abnormally last time the specified user executed it.

***YES**

Run the task's recovery program if the task terminated abnormally last time the specified user executed it.

This parameter is ignored in the following circumstances:

- The Verify Allocation routine has been run since the task last terminated abnormally.
- There is no recovery program specified for the task.

Prompt Before Submitting (PROMPT)

***NO**

Do not display a screen which allows the user to modify the job submission parameters.

***YES**

Display a screen which allow the user to modify the job submission details.

This parameter is ignored in the following circumstances:

- The command is being executed within a batch environment.
- The user is not permitted to modify job submission parameters. (This is determined by examining the user's profile set up in Administration Functions.)
- The task is interactive only.

Restore System Libraries (RSTSYSLIBL)

***NO**

When the command has completed, the system library list will not retain any Application libraries. Use *NO when the system library list does not contain Application libraries to start with. This will usually be the case if the command is being run outside Application Manager.

***YES**

Extra work is done to ensure the system library list is left in it's original condition. Use *YES when the system library list contains some Application Manager libraries even though the command is being used outside Application Manager.

***ASSUME**

This option involves the least extra processing, but it may only be used when already within Application Manager, that is, when the system library list has already been set for use with Application Manager.